

Mitsumasa Koyanagi

professor

基本信息

姓名: Mitsumasa Koyanagi

所属院校: 东北大学 (仙台)

地区: Japan

荣誉称号

★ IEEE Fellow

这些成就代表了在学术领域的最高认可，展现了卓越的研究能力和影响力。

学科领域

微电子科学与工程

集成电路设计与集成系统

研究领域

微电子器件与制造

微电子封装与可靠性

纳米合成与组装技术

纳米电子与先进工艺

纳米电子学与光子学

超大规模集成电路设计

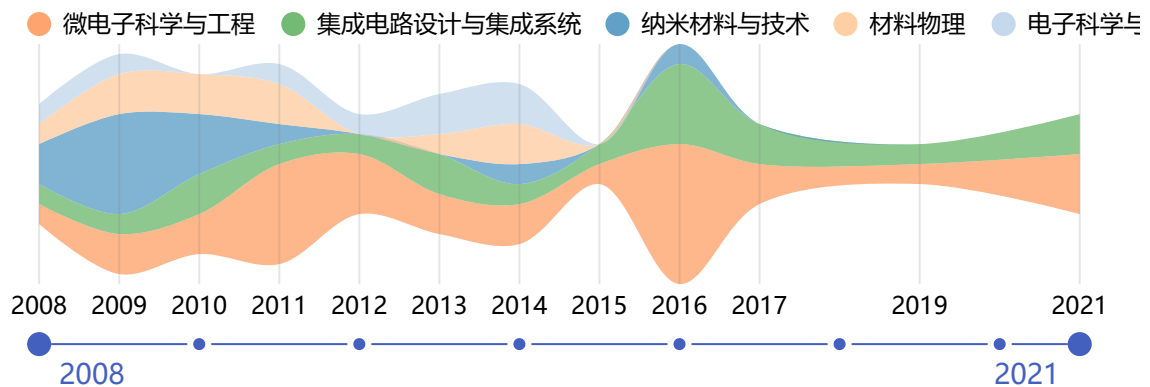
混合信号集成技术

个人简介

Mitsumasa Koyanagi is a professor in the Department of Bioengineering and Robotics at Tohoku University. His fields of study include memory. He, along with Hideo Sunami and Kiyoo Itoh, made significant contributions to modern DRAM development, devising trench, stacked capacitor, and folded data line cells. In 1976, he devised the stacked capacity cell. His work has been applied to various memory devices. He also pioneered 3D integrated circuit fabrication. An IEEE Fellow, he has received numerous awards, such as the IEEE

Cledo Brunetti Award, SSDM Award, Okochi Memorial Technology Prize, and IEEE Electronics Packaging Award.

近年发表的论文



Design for 3-D Stacked Neural Network Circuit with Cyclic Analog Computing

2021 IEEE INTERNATIONAL 3D SYSTEMS INTEGRATION CONFERENCE (3DIC)

Kiyoyama, Koji; Horio, Yoshihiko; Fukushima, Takafumi; Hashimoto, Hiroyuki; Orima, Takemori; Koyanagi, Mitsumasa

被引用次数: 2

2021

Implementation of a Chaotic Neural Network Reservoir on a TSV/ μ Bump Stacked 3D Cyclic Neural Network Integrated Circuit

2021 IEEE INTERNATIONAL 3D SYSTEMS INTEGRATION CONFERENCE (3DIC)

Horio, Yoshihiko; Orima, Takemori; Kiyoyama, Koji; Koyanagi, Mitsumasa

被引用次数: 0

2021

Integration of Damage-less Probe Cards Using Nano-TSV Technology for Microbumped Wafer Testing

2021 IEEE INTERNATIONAL 3D SYSTEMS INTEGRATION CONFERENCE (3DIC)

Fukushima, Takafumi; Sakuyama, Shinichi; Takahashi, Masatomo; Hashimoto, Hiroyuki; Bea, Jichael; Marcello, Theodorus; Kino, Hisashi; Tanaka, Tetsu; Koyanagi, Mitsumasa; Mariappan, Murugesan

被引用次数: 7

2021

On-wafer thermomechanical characterization of a thin film polyimide formed by vapor deposition polymerization through-silicon via applications: Comparison to plasma-enhanced chemical vapor deposition SiO₂

JOURNAL OF POLYMER SCIENCE

Fukushima, Takafumi; Murugesan, Mariappan; Bea, Ji-Cheol; Hashimoto, Hiroyuki; Kino, Hisashi; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 3

2020

High-Thermoresistant Temporary Bonding Technology for Multichip-to-Wafer 3-D Integration With Via-Last TSVs

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Hashiguchi, Hideto; Fukushima, Takafumi; Murugesan, Mariappan; Kino, Hisashi; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 7

2019

Self-Assembly and Electrostatic Carrier Technology for Via-Last TSV Formation Using Transfer Stacking-Based Chip-to-Wafer 3-D Integration

IEEE TRANSACTIONS ON ELECTRON DEVICES

Hashiguchi, Hideto; Fukushima, Takafumi; Hashimoto, Hiroyuki; Bea, Ji-Cheol; Murugesan, Mariappan; Kino, Hisashi; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 12

2017

3-D Sidewall Interconnect Formation Climbing Over Self-Assembled KGDs for Large-Area Heterogeneous Integration

IEEE TRANSACTIONS ON ELECTRON DEVICES

Fukushima, Takafumi; Noriki, Akihiro; Bea, Jichoel; Murugesan, Mariappan; Kino, Hisashi; Kiyoyama, Koji; Lee, Kang-Wook; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 6

2017

Evaluation of in-plane local stress distribution in stacked IC chip using dynamic random access memory cell array for highly reliable three-dimensional IC

JAPANESE JOURNAL OF APPLIED PHYSICS

Tanikawa, Seiya; Kino, Hisashi; Fukushima, Takafumi; Koyanagi, Mitsumasa; Tanaka, Tetsu

被引用次数: 0

2016

Novel Hybrid Bonding Technology Using Ultra-High Density Cu Nano-Pillar for Exascale 2.5D/3D Integration

IEEE ELECTRON DEVICE LETTERS

Lee, Kangwook; Bea, Jichel; Fukushima, Takafumi; Ramalingam, Suresh; Wu, Xin; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 11

2016

Effect of local stress induced by thermal expansion of underfill in three-dimensional stacked IC

JAPANESE JOURNAL OF APPLIED PHYSICS

Kino, Hisashi; Hashiguchi, Hideto; Tanikawa, Seiya; Sugawara, Youhei; Ikegaya, Shunsuke; Fukushima, Takafumi; Koyanagi, Mitsumasa; Tanaka, Tetsu

被引用次数: 7

2016

Heterogeneous 3-D Integration Using Self-Assembly and Electrostatic Bonding

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Koyanagi, Mitsumasa; Fukushima, Takafumi; Lee, Kang-Wook; Tanaka, Tetsu

被引用次数: 5

2016

Impact of Chip-Edge Structures on Alignment Accuracies of Self-Assembled Dies for Microelectronic System Integration

JOURNAL OF MICROELECTROMECHANICAL SYSTEMS

Ito, Yuka; Fukushima, Takafumi; Kino, Hisashi; Lee, Kang-Wook; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 15

2016

Study of Vacuum-Assisted Spin Coating of Polymer Liner for High-Aspect-Ratio Through-Silicon-Via Applications

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Yan, Yangyang; Ding, Yingtao; Fukushima, Takafumi; Lee, Kang-Wook; Koyanagi, Mitsumasa

被引用次数: 12

2016

Oxide-Oxide Thermocompression Direct Bonding Technologies with Capillary Self-Assembly for Multichip-to-Wafer Heterogeneous 3D System Integration

MICROMACHINES

Fukushima, Takafumi; Hashiguchi, Hideto; Yonekura, Hiroshi; Kino, Hisashi; Murugesan, Mariappan; Bea, Ji-Chel; Lee, Kang-Wook; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 21

2016

Capacitance characteristics of low-k low-cost CVD grown polyimide liner for high-density Cu through-Si-via in three-dimensional LSI

JAPANESE JOURNAL OF APPLIED PHYSICS

Mariappan, Murugesan; Fukushima, Takafumi; Bea, Ji-Chel; Hashimoto, Hiroyuki; Koyanagi, Mitsumasa

被引用次数: 6

2016

Applications of three-dimensional LSI

MRS BULLETIN

Koyanagi, Mitsumasa; Fukushima, Takafumi; Lee, Kang-Wook; Tanaka, Tetsu

被引用次数: 9

2015

Mechanical Characteristics of Thin Die/Wafers in Three-Dimensional Large-Scale Integrated Systems

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Mariappan, Murugesan; Fukushima, Takafumi; Bea, Jichael C.; Lee, Kang-Wook; Koyanagi, Mitsumasa

被引用次数: 9

2014

Impacts of 3-D Integration Processes on Memory Retention Characteristics in Thinned DRAM Chip for High-Reliable 3-D DRAM

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lee, Kang-Wook; Tanikawa, Seiya; Murugesan, Mariappan; Naganuma, Hideki; Bea, Ji-Chael; Fukushima, Takafumi; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 13

2014

Deteriorated Device Characteristics in 3D-LSI Caused by Distorted Silicon Lattice

IEEE TRANSACTIONS ON ELECTRON DEVICES

Mariappan, Murugesan; Imai, Yasuhiko; Kimura, Shigeru; Fukushima, Takafumi; Bea, Ji-Chael; Kino, Hisashi; Lee, Kang-Wook; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 4

2014

Impacts of Cu Contamination on Device Reliabilities in 3-D IC Integration

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Lee, Kang-Wook; Bea, Ji-Chel; Ohara, Yuki; Murugesan, Mariappan; Fukushima, Takafumi; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 16

2014

Reconfigured-Wafer-to-Wafer 3-D Integration Using Parallel Self-Assembly of Chips With Cu-SnAg Microbumps and a Nonconductive Film

IEEE TRANSACTIONS ON ELECTRON DEVICES

Fukushima, Takafumi; Bea, Jichael; Kino, Hisashi; Nagai, Chisato; Murugesan, Mariappan; Hashiguchi, Hideto; Lee, Kang-Wook; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 45

2014

Barrier Properties of CVD Mn Oxide Layer to Cu Diffusion for 3-D TSV

IEEE ELECTRON DEVICE LETTERS

Lee, Kang-Wook; Wang, Hao; Bea, Ji-Cheol; Murugesan, Mariappan; Sutou, Yuji; Fukushima, Takafumi; Tanaka, Tetsu; Koike, Junichi; Koyanagi, Mitsumasa

被引用次数: 17

2014

Die-Level 3-D Integration Technology for Rapid Prototyping of High-Performance Multifunctionality Hetero-Integrated Systems

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lee, Kang-Wook; Ohara, Yuki; Kiyoyama, Kouji; Bea, Ji-Cheol; Murugesan, Mariappan; Fukushima, Takafumi; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 22

2013

Degradation of Memory Retention Characteristics in DRAM Chip by Si Thinning for 3-D Integration

IEEE ELECTRON DEVICE LETTERS

Lee, Kangwook; Tanikawa, Seiya; Murugesan, Mariappine; Naganuma, Hideki; Shimamoto, Haro; Fukushima, Takafumi; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 21

2013

Investigation of Local Bending Stress Effect on Complementary Metal-Oxide-Semiconductor Characteristics in Thinned Si Chip for Chip-to-Wafer Three-Dimensional Integration

JAPANESE JOURNAL OF APPLIED PHYSICS

Kino, Hisashi; Bea, Ji Choel; Murugesan, Mariappan; Lee, Kang Wook; Fukushima, Takafumi; Koyanagi, Mitsumasa; Tanaka, Tetsu

被引用次数: 8

2013

Reductant-Assisted Self-Assembly with Cu/Sn Microbump for Three-Dimensional Heterogeneous Integration

JAPANESE JOURNAL OF APPLIED PHYSICS

Ito, Yuka; Fukushima, Takafumi; Lee, Kang-Wook; Choki, Koji; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 11

2013

Impact of Cu Contamination on Memory Retention Characteristics in Thinned DRAM Chip for 3-D Integration

IEEE ELECTRON DEVICE LETTERS

Lee, Kangwook; Tani, Takaharu; Naganuma, Hideki; Ohara, Yuki; Fukushima, Takafumi; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 11

2012

Through-Silicon Photonic Via and Unidirectional Coupler for High-Speed Data Transmission in Optoelectronic Three-Dimensional LSI

IEEE ELECTRON DEVICE LETTERS

Noriki, Akihiro; Lee, Kangwook; Bea, Jicheol; Fukushima, Takafumi; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 21

2012

Multichip-to-Wafer Three-Dimensional Integration Technology Using Chip Self-Assembly With Excimer Lamp Irradiation

IEEE TRANSACTIONS ON ELECTRON DEVICES

Fukushima, Takafumi; Iwata, Eiji; Ohara, Yuki; Murugesan, Mariappan; Bea, Jichoel; Lee, Kangwook; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 45

2012

Three-Dimensional Hybrid Integration Technology of CMOS, MEMS, and Photonics Circuits for Optoelectronic Heterogeneous Integrated Systems

IEEE TRANSACTIONS ON ELECTRON DEVICES

Lee, Kang-Wook; Noriki, Akihiro; Kiyoyama, Kouji; Fukushima, Takafumi; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 96

2011

Evaluation of Cu Diffusion From Cu Through-Silicon Via (TSV) in Three-Dimensional LSI by Transient Capacitance Measurement

IEEE ELECTRON DEVICE LETTERS

Bea, Jichel; Lee, Kangwook; Fukushima, Takafumi; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 40

2011

Multichip Self-Assembly Technology for Advanced Die-to-Wafer 3-D Integration to Precisely Align Known Good Dies in Batch Processing

IEEE TRANSACTIONS ON COMPONENTS PACKAGING AND MANUFACTURING TECHNOLOGY

Fukushima, Takafumi; Iwata, Eiji; Ohara, Yuki; Murugesan, Mariappan; Bea, Jichoel; Lee, Kangwook; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 40

2011

Self-Assembly of Chip-Size Components with Cavity Structures: High-Precision Alignment and Direct Bonding without Thermal Compression for Hetero Integration

MICROMACHINES

Fukushima, Takafumi; Konno, Takayuki; Iwata, Eiji; Kobayashi, Risato; Kojima, Toshiya; Murugesan, Mariappan; Bea, Ji-Chel; Lee, Kang-Wook; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 25

2011

MOSFET Nonvolatile Memory with High-Density Cobalt-Nanodots Floating Gate and HfO₂ High-k Blocking Dielectric

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Pei, Yanli; Yin, Chengkuan; Kojima, Toshiya; Bea, Ji-Cheol; Kino, Hisashi; Fukushima, Takafumi; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 30

2011

Multilevel Charge Storage in a Multiple Alloy Nanodot Memory

JAPANESE JOURNAL OF APPLIED PHYSICS

Lee, Gae-Hun; Lee, Jung-Min; Song, Yun Heub; Bea, Ji Chel; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 6

2011

Evaluation of Cu Contamination at Backside Surface of Thinned Wafer in 3-D Integration by Transient-Capacitance Measurement

IEEE ELECTRON DEVICE LETTERS

Bea, Jichel; Lee, Kangwook; Fukushima, Takafumi; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 29

2011

Novel interconnection technology for heterogeneous integration of MEMS-LSI multi-chip module

MICROSYSTEM TECHNOLOGIES-MICRO-AND NANOSYSTEMS-INFORMATION STORAGE AND PROCESSING SYSTEMS

Lee, Kang-Wook; Koyanagi, Mitsumasa

被引用次数: 2

2010

Effects of Postdeposition Annealing on Cobalt Nanodots Embedded in Silica for Nonvolatile Memory Application

JAPANESE JOURNAL OF APPLIED PHYSICS

Pei, Yanli; Kojima, Toshiya; Hiraki, Tatsuro; Fukushima, Takafumi; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 2

2010

Characteristics of a Multiple Alloy Nanodot Memory with an Enhanced Charge Storage Capability

JAPANESE JOURNAL OF APPLIED PHYSICS

Song, Yun Heub; Bea, Ji Chel; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 3

2010

A Cavity Chip Interconnection Technology for Thick MEMS Chip Integration in MEMS-LSI Multichip Module

JOURNAL OF MICROELECTROMECHANICAL SYSTEMS

Lee, Kang-Wook; Kanno, Soichiro; Kiyoyama, Kouji; Fukushima, Takafumi; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 23

2010

A Reliable Nonvolatile Memory Using Alloy Nanodot Layer with Extremely High Density (vol 48, 106505, 2009)

JAPANESE JOURNAL OF APPLIED PHYSICS

Song, Yun Heub; Bea, Ji Chel; Lee, Kang Wook; Lee, Gae-Hun; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 0

2009

A Reliable Nonvolatile Memory Using Alloy Nanodot Layer with Extremely High Density

JAPANESE JOURNAL OF APPLIED PHYSICS

Song, Yun Heub; Bea, Ji Chel; Lee, Kang Wook; Lee, Gae-Hun; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 0

2009

Formation of high density tungsten nanodots embedded in silicon nitride for nonvolatile memory application

APPLIED PHYSICS LETTERS

Pei, Yanli; Yin, Chengkuan; Nishijima, Masahiko; Kojima, Toshiya; Fukushima, Takafumi; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 9

2009

Cell characteristics of a multiple alloy nano-dots memory structure

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Bea, Ji Chel; Song, Yun Heub; Lee, Kang-Wook; Lee, Gae-Hun; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 9

2009

Characteristics of Copper Spiral Inductors Utilizing FePt Nanodot Films

JAPANESE JOURNAL OF APPLIED PHYSICS

Jeong, Woo-Cheol; Kiyoyama, Kouji; Lee, Kang-Wook; Noriki, Akihiro; Murugesan, Mariappan; Fukushima, Takafumi; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 1

2009

Fundamental Study of Complementary Metal Oxide Semiconductor Image Sensor for Three-Dimensional Image Processing System

JAPANESE JOURNAL OF APPLIED PHYSICS

Makita, Kenji; Kiyoyama, Kouji; Sugimura, Takeaki; Lee, Kang Wook; Fukushima, Takafumi; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 6

2009

High-Density Through Silicon Vias for 3-D LSIs

PROCEEDINGS OF THE IEEE

Koyanagi, Mitsumasa; Fukushima, Takafumi; Tanaka, Tetsu

被引用次数: 258

2009

Memory characteristics of metal-oxide-semiconductor capacitor with high density cobalt nanodots floating gate and HfO₂ blocking dielectric

APPLIED PHYSICS LETTERS

Pei, Yanli; Yin, Chengkuan; Kojima, Toshiya; Nishijima, Masahiko; Fukushima, Takafumi; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 19

2009

Electrical characterization of metal-oxide-semiconductor memory devices with high-density self-assembled tungsten nanodots

JAPANESE JOURNAL OF APPLIED PHYSICS

Pei, Yan-Li; Fukushima, Takafumi; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 162008

New reconfigurable memory architecture for parallel image-processing LSI with three-dimensional structure

JAPANESE JOURNAL OF APPLIED PHYSICS

Kodama, Shigeo; Amano, Daijiro; Sugimura, Takeaki; Fukushima, Takafumi; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 32008

Tungsten through-silicon via technology for three-dimensional LSIs

JAPANESE JOURNAL OF APPLIED PHYSICS

Kikuchi, Hirokazu; Yamada, Yusuke; Ali, Atif Mossad; Liang, Jun; Fukushima, Takafumi; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 582008

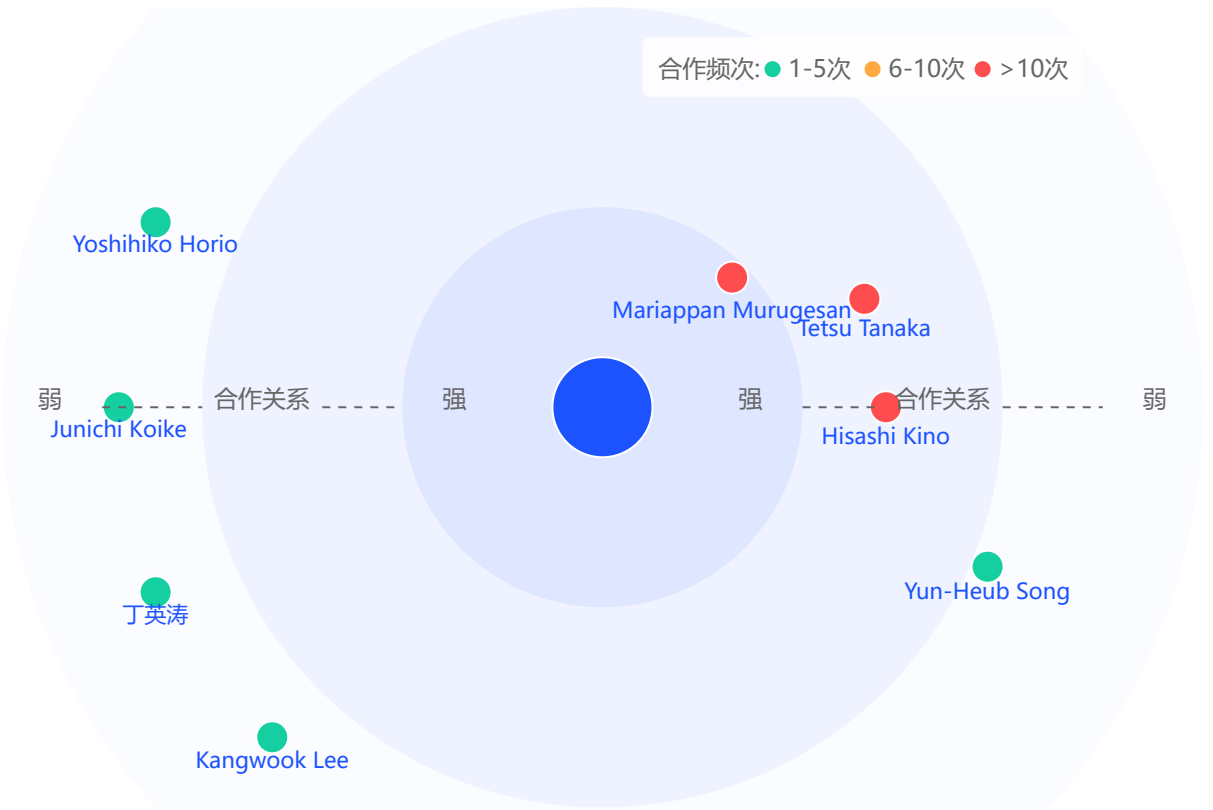
Memory characteristics of self-assembled tungsten nanodots dispersed in silicon nitride

APPLIED PHYSICS LETTERS

Pei, Yanli; Nishijima, Masahiko; Fukushima, Takafumi; Tanaka, Tetsu; Koyanagi, Mitsumasa

被引用次数: 142008

学术合作



Mariappan Murugesan 东北大学（仙台）	合作次数：19
Tetsu Tanaka 东北大学（仙台）	合作次数：13
Hisashi Kino 东北大学（仙台）	合作次数：13
Yun-Heub Song 汉阳大学	合作次数：5
Kangwook Lee 东北大学（仙台）	合作次数：2

还有 3 个合作者